

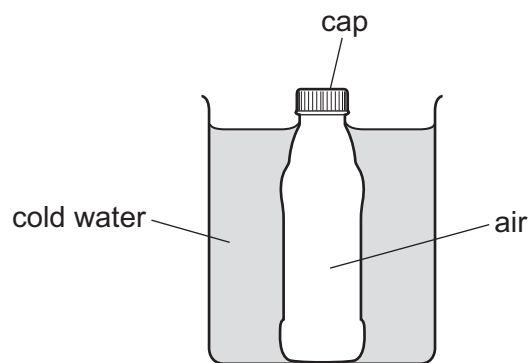
- 10 Which situation is **not** explained by the equation $P = \frac{F}{A}$?
- A using a longer spanner than normal to undo a tight nut
 - B hammering a nail into a piece of wood
 - C tractors using wide tyres in a muddy field
 - D a sharp kitchen knife cutting vegetables more easily than a blunt one

- 11 A substance is heated. The substance changes state from a liquid to a gas.

When the gas is cooled, it first becomes a liquid and finally a solid.

What is the order of changes of state for this substance?

- A condensation → melting → freezing
 - B vaporisation → condensation → melting
 - C vaporisation → condensation → freezing
 - D melting → vaporisation → freezing
- 12 A glass bottle containing warm air is sealed with a screw cap and then cooled in cold water.



The contraction of the glass bottle can be ignored.

What remains the same during the cooling?

- A the air pressure inside the bottle
- B the energy of the air particles in the bottle
- C the force on the cap made by the air particles in the bottle
- D the volume of air in the bottle